Phishing Awareness Training

- CodeAlpha Cyber Security Internship – Task 2
- Recognize, resist and report phishing.
- All brands and domains in this deck are fictional.

1. Introduction to Phishing

- Phishing tricks people into revealing credentials, paying money or running malware.
- It exploits human trust more than technical flaws.
- Channels: email, SMS, voice calls, fake websites.

2. How Phishing Attacks Work

- Bait: a convincing message.
- Hook: link, attachment or request.
- Capture: credentials, payment or malware.
- Abuse: account takeover, fraud, lateral attacks.

3. Types of Phishing

- Email phishing – mass lures posing as a brand.
- Spear phishing – targeted, personalised.
- Whaling – aimed at executives.
- Smishing – SMS. Vishing – voice calls.
- Clone phishing – a real email copied with a malicious link.

4. Social Engineering Techniques

- Urgency and fear – deadlines and threats.
- Authority and impersonation – boss, bank, IT.
- Trust and familiarity – known names and logos.
- Curiosity and rewards – prizes, refunds, 'see this file'.

5. Identifying Suspicious Emails

- Sender domain does not match the organisation.
- Generic greeting, odd tone, spelling errors.
- Unexpected request, urgency or secrecy.
- Links whose preview differs from the text.

6. Identifying Fake Websites

- Check the domain before typing credentials.
- A padlock means encrypted, not genuine.
- Be wary of pages reached through email links.
- Unexpected requests for PIN, card or MFA codes.

7. URL Analysis and Domain Spoofing

- Real domain = the part just before the first single '/'.
- Example: northbridge.example.evil-site.example belongs to evil-site.example.
- Typosquatting: cloud1y.example vs cloudly.example.
- Extra words: northbridge-secure-login.example is not northbridge.example.

8. Malicious Attachments and Links

- Unexpected invoices, documents and archives.
- Double extensions such as .pdf.exe.
- Macros and 'enable content' prompts.
- Verify through a separate channel before opening.

9. Realistic Scenarios (fictional)

- Account suspension notice from 'Northbridge Bank'.
- Parcel fee SMS from 'Parcelo Express'.
- CEO gift-card request (business email compromise pattern).
- Bank 'fraud team' call asking for PIN.

10. Prevention Best Practices

- Use unique passwords and a password manager.
- Enable MFA; prefer passkeys or security keys.
- Keep software updated.
- Verify unusual requests out-of-band.
- Report suspicious messages.

11. After Clicking a Phishing Link

- Disconnect if you opened a file.
- Change passwords from a trusted device.
- Enable/reset MFA, review account activity.
- Report to IT/security.
- Scan the device.

12. Final Knowledge Check

- Take the 10-question interactive quiz in interactive_module/.
- Review explanations for any missed answers.
- Sources: see references/sources.md (CISA, NIST, Canadian Centre for Cyber Security, Microsoft).